gaining a match between a face on the database and the more recently collected image.[2] FRT's commercial applications range from enabling more effective photo sharing on social media sites by identifying faces on images uploaded to platforms such as Facebook and Instagram and linking these to user profiles, through to authenticating employees to access secure premises such as a power plant or prison.[3] Buolamwini and Gebru note that the capacity of FRT to move beyond mere face detection and towards the identification of emotions and personality characteristics is also increasing rapidly and becoming more precise (although such developments remain in their infancy).[4] FRT even holds the potential to ascertain a person's sexuality.[5]

Advances in FRT also have numerous criminal justice and policing applications, and this technology is becoming increasingly popular for police forces across the world. Where successful, such applications often have headline-grabbing effects.[6] FRT has been trialled by a handful of domestic police forces looking to position themselves at the leading edge of technology-led crime control practice. In England and Wales, FRT cameras have been used predominantly in the context of public surveillance operations at large gatherings such as outdoor festivals, sports events or public protests.[7] Though other applications of FRT may be utilised by the police in the future, which would undoubtedly raise significant practical and principled issues, these will not be considered here. In the interests of developing sharper focus, the following analysis is

[2] L.D. Introna and H. Nissenbaum, *Facial Recognition Technology: A Survey of Policy and Implementation Issues* (Center for Catastrophe Preparedness and Response, 2009), p.11.
[3] A.P. Cackley, *Facial Recognition Technology: Commercial Uses, Privacy Issues, and Applicable Federal Law* (US Government Accountability Office, 2015), p.3.
[4] J. Buolamwini and T. Gebru, "Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification" (Conference on Fairness, Accountability, and Transparency, New York, NY, 2018), p.2.
[5] Y. Wang and M. Kosinski, "Deep neural networks are more accurate than humans at detecting sexual orientation from facial images" (2018) 114 J. Pers. Soc. Psychol. 246.
[6] Recently, police in India suggested that the roll out of FRT across New Delhi enabled them to identify 3,000 missing children in just four days. See A. Cuthbertson, "Indian Police trace 3,000 Missing Children In Just Four Days Using Facial Recognition Technology" (April 24, 2018), *Independent.co.uk*, *https://www.independent.co.uk/life-style/gadgets-and-tech/news/india-police-missi ng-children-facial-recognition-tech-trace-find-reunite-a8320406.html*; "Chinese man caught by facial recognition at pop concert" (April 13, 2018), *BBC.co.uk*, *http://www.bbc.co.uk/news/world-asia-china-43751276* [Accessed August 7, 2018].
[7] "Three arrested using facial recognition technology during Wales' Six Nations opener" (February 6, 2018), *Walesonline.co.uk*, *https://www.walesonline.co.uk/news/wales-news/three-arrested-using-facial-recognition-14253344* [Accessed August 4, 2018]; V. Dodd, "Met police to use facial recognition software at Notting Hill carnival" (August 5, 2017), *Theguardian.co.uk*, *https://www.theguardian.com/uk-news/2017/aug/05/met-police-facial-recognition-software-notting-hill-carnival* [Accessed August 7, 2018].

The MPS first used FRT at the Notting Hill Carnival in 2016, and aims to complete ten trials of its system by the end of 2018.[10] It has a bespoke facial recognition system. The trial in 2016 resulted in no successful identifications, and just one successful identification at the 2017 Notting Hill Carnival; in 2017 the system also misidentified five carnival goers as constituting wanted individuals, who were subject to a brief stop by police as a result of the "false positive" match.[11] These individuals were subject to stops despite the efforts of the MPS to reduce the risk of misidentification by adopting a two-step verification process, whereby any matches are checked by a human operator before being passed to patrolling officers in the vicinity of the monitored public space.[12] The MPS trialled their system for a third time on Remembrance Sunday 2017, again making one positive identification to a person on a watch list. This trial was particularly controversial as the MPS compiled and used a watch list of "fixated individuals", who were identified as having obsessive tendencies towards certain public figures but were not wanted in connection with any specific offence. According to *The Independent*, the MPS's use of the technology in 2018 in Stratford yielded no arrests.[13]

SWP is the national lead on FRT, having received a £2.6 million Government grant to test the technology.[14] SWP deployed Neoface's FRT system at 18 public gatherings between May 2017 and March 2018, and has no set end date for its trial of FRT. Big Brother Watch raised concerns about the accuracy of the technology during these trials, noting that a "staggering 91% of matches—2,451—incorrectly identified innocent members of the public."[15] SWP has defended its continued use of FRT, publishing

[10] Big Brother Watch, *Face Off: The lawless growth of facial recognition in UK policing* (London, 2018), p.26.
[11] A false positive match occurs where a FRT system mistakenly matches a person passing under a facial recognition camera to an image on the watch list. Big Brother Watch, *Face Off: The lawless growth of facial recognition in UK policing*, p.26.
[12] Facial identification is most accurate when both algorithms and so called "super-recognisers" (humans who are particularly adept at face identification) work in collaboration: see P. Phillips et al, "Face recognition accuracy of forensic examiners, super-recognizers, and face recognition algorithms" (2018) 115 Proc. Natl. Acad. Sci. USA. 6171. However, there is no legal prerequisite for the MPS to use super-recognisers when crosschecking FRT matches, and this is not common practice: see G. Edmond and N. Wortley, "Interpreting Image Evidence: Facial Mapping, Police Familiars and Super-recognisers in England and Australia" (2016) 3 J.I.C.L. 473.
[13] "Facial recognition trial in London results in zero arrests, Metropolitan Police confirm" (July 3, 2018) *Independent.co.uk*, *https://www.independent.co.uk/news/uk/crime/facial-recognition-police-uk-london-trials-stratford-no-arrests-privacy-human-rights-false-positives-a8429466.html* [Accessed August 1, 2018].
[14] *https://www.gov.uk/government/publications/police-transformation-fund-successful-bids-2016-to-2017* [Accessed August 1, 2018].
[15] Big Brother Watch, *Face Off: The lawless growth of facial recognition in UK policing*, p.29.